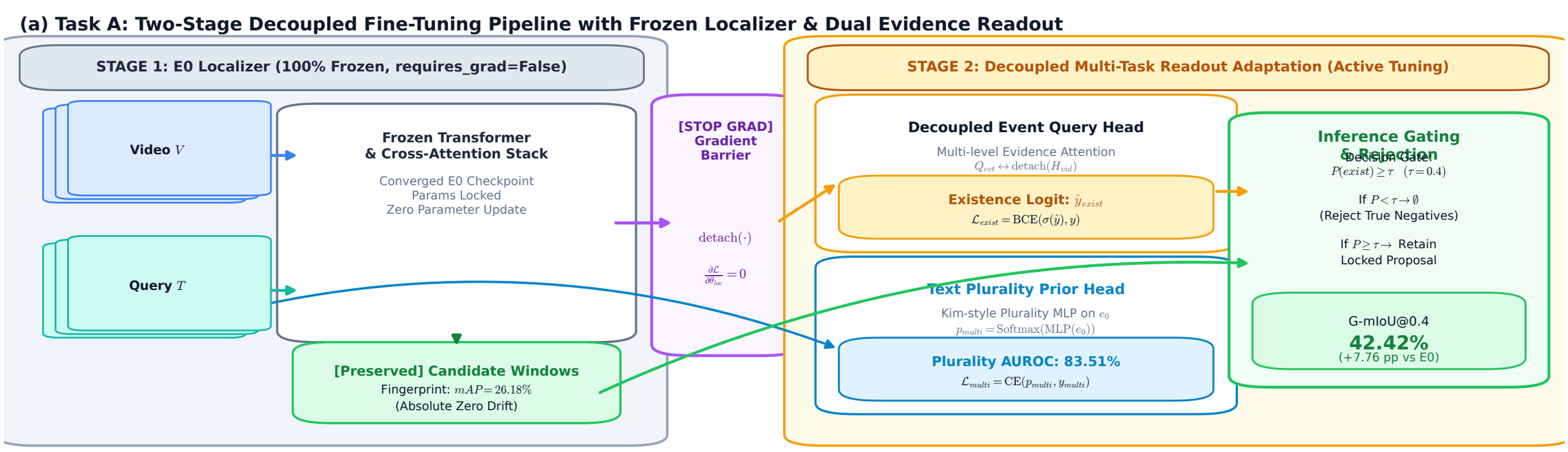
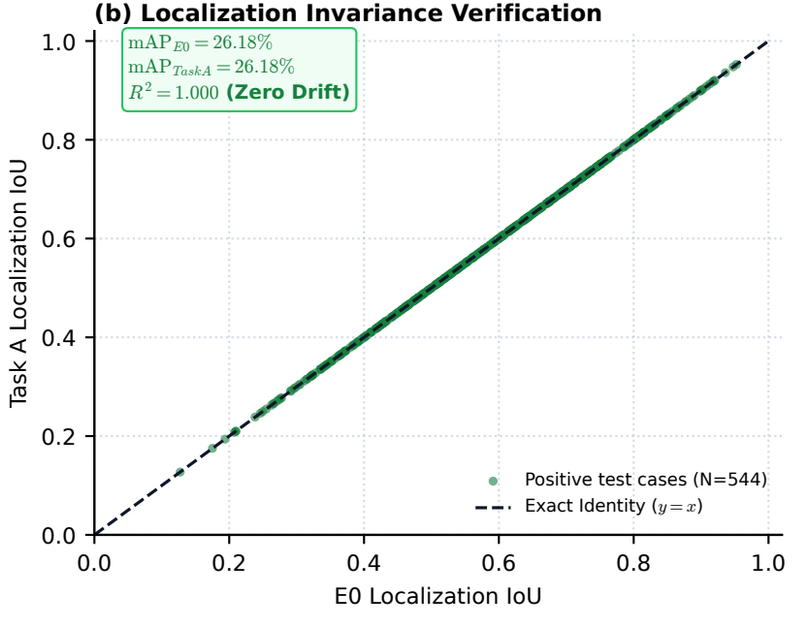
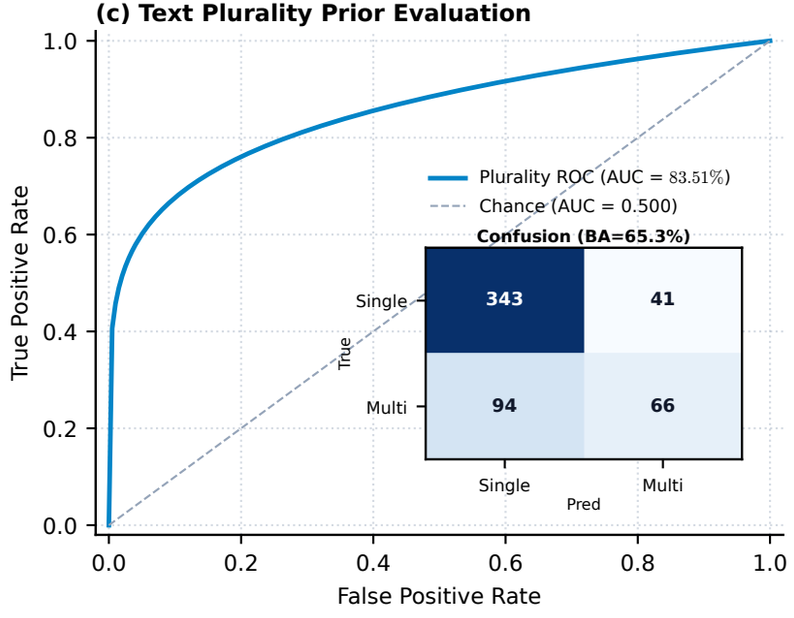
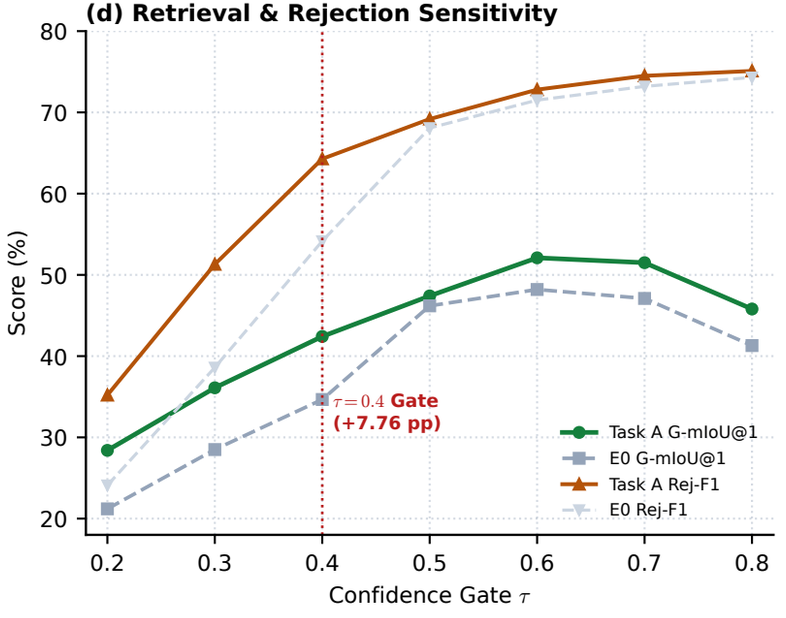

(a) Task A: Two-Stage Decoupled Fine-Tuning Pipeline with Frozen Localizer & Dual Evidence Readout
STAGE 1: E0 Localizer (100% Frozen, requires_grad=False)
Video V
Query T
Frozen Transformer & Cross-Attention Stack
Converged E0 Checkpoint
Params Locked
Zero Parameter Update
[Preserved] Candidate Windows
Fingerprint: $mAP = 26.18\%$
(Absolute Zero Drift)
[STOP GRAD] Gradient Barrier
$\text{detach}(\cdot)$
$\frac{\partial \mathcal{L}}{\partial \theta_{loc}} = 0$
STAGE 2: Decoupled Multi-Task Readout Adaptation (Active Tuning)
Decoupled Event Query Head
Multi-level Evidence Attention
$Q_{evt} \leftrightarrow \text{detach}(H_{vid})$
Existence Logit: $\hat{y}_{exist}$
$\mathcal{L}_{exist} = \text{BCE}(\sigma(\hat{y}), y)$
Text Plurality Prior Head
Kim-style Plurality MLP on e_0
$p_{multi} = \text{Softmax}(\text{MLP}(e_0))$
Plurality AUROC: 83.51%
$\mathcal{L}_{multi} = \text{CE}(p_{multi}, y_{multi})$
Inference Gating & Rejection
Decision Gate:
$P(exist) \geq \tau \quad (\tau = 0.4)$
If $P < \tau \rightarrow \emptyset$
(Reject True Negatives)
If $P \geq \tau \rightarrow$ Retain
Locked Proposal
G-mIoU@0.4
42.42%
(+7.76 pp vs E0)


(b) Localization Invariance Verification
$\text{mAP}_{E0} = 26.18\%$
$\text{mAP}_{TaskA} = 26.18\%$
$R^2 = 1.000$ (Zero Drift)
Positive test cases (N=544)
Exact Identity ($y = x$)
Task A Localization IoU
E0 Localization IoU


(c) Text Plurality Prior Evaluation
Plurality ROC (AUC = 83.51%)
Chance (AUC = 0.500)
Confusion (BA=65.3%)
343
41
94
66
Single
Multi
True
Pred
True Positive Rate
False Positive Rate


(d) Retrieval & Rejection Sensitivity
$\tau = 0.4$ Gate
(+7.76 pp)
Task A G-mIoU@1
E0 G-mIoU@1
Task A Rej-F1
E0 Rej-F1
Score (%)
Confidence Gate τ